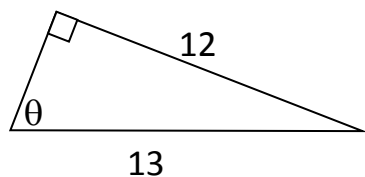


- 1) Use the triangle below to write the six trigonometric ratios of angle θ . Give your answer in reduced fraction form. (no decimals)



$$\sin \theta = \underline{\hspace{2cm}}$$

$$\cot \theta = \underline{\hspace{2cm}}$$

$$\cos \theta = \underline{\hspace{2cm}}$$

$$\sec \theta = \underline{\hspace{2cm}}$$

$$\tan \theta = \underline{\hspace{2cm}}$$

$$\csc \theta = \underline{\hspace{2cm}}$$

- 2) Find the five remaining trig functions given the $\csc \theta = 2$

On #3 and 4, determine the quadrant in which the angle lies.

On #5 and 6, sketch the angle in standard position.

Then for each of #3 – 6, list one positive and one negative co-terminal angle.

3. $-\frac{37\pi}{24}$

4. $\frac{5\pi}{7}$

5. $\frac{2\pi}{9}$

6. $-\frac{7\pi}{4}$

For #7-10, find the complement and supplement (if possible) of the following angles.

7. $\frac{\pi}{8}$

8. $\frac{13\pi}{12}$

9. $\frac{3\pi}{10}$

10. $\frac{2\pi}{21}$

On #11 and 12, determine the quadrant in which the angle lies.

On #13 and 14, sketch the angle in standard position.

Then for each of #11 – 14, list one positive and one negative co-terminal angle.

11. 245°

12. -405°

13. -130°

14. 210°

Find the complement (if possible) and supplement of the angle.

15. 5°

16. 184°

17. 71°

18. 136°

Convert the angle measures from degrees to radians.

19. 415°

20. -335°

21. -72°

22. 94°

Convert the angle measures from radians to degrees.

23. $\frac{5\pi}{7}$

24. $-\frac{3\pi}{5}$

25. $-\frac{7\pi}{6}$

26. $\frac{5\pi}{4}$

27. A neighborhood carnival has a Ferris wheel whose radius is 30 feet. You measure the time it takes for one revolution to be 70 seconds. What is the linear speed (in feet per second) of this Ferris wheel? What is the angular speed in radians per second?

28. The second hand on Mr. Incredible's watch is 0.25 inches long. How fast is the tip of the second hand moving?

29. To approximate the speed of the current of a river, a circular paddle wheel with diameter 8 feet is lowered into the water. If the current causes the wheel to rotate at a speed of 10 revolutions per minute, what is the speed of the current? Express your answer in miles per hour.